ENVIRONMENTAL STUDIES

Resource Limits, Environmental Stress & Sustainability

Carrying Capacity and Human–Environment Interactions

Module I | S.Y. B.Tech Civil Engineering

“Development depends on nature — but nature also has limits.”

02

From Human Pressure to Environmental Limits

Population Growth → Urbanization → Industrialization → Infrastructure Expansion → Resource Demand

What happens when resource demand approaches environmental capacity?

Resource Pressure → Environmental Stress → Depletion / Degradation → Reduced Capacity

Engineering question: Can demand be reduced or redesigned before environmental limits are crossed?

Renewable and Non-Renewable Resources

RENEWABLE / REPLENISHABLE

Naturally replenished / regenerated within a human-relevant timescale.

Solar radiation • Wind • Tidal energy • Water • Forests • Biomass

NON-RENEWABLE / EXHAUSTIBLE

Finite geological stocks; not replenished within a human timescale.

Coal • Petroleum • Natural gas • Iron ore • Copper • Limestone

Key distinction: renewable does not mean unlimited.

Distinguish renewable ENERGY sources from renewable NATURAL resources.

Renewable Does Not Mean Unlimited

Sustainability depends on the balance between regeneration and human demand.

Regeneration > Use	Regeneration ≈ Use	Regeneration < Use
Sustainable use	Stress / balance	Depletion

Groundwater • Forests • Fisheries • Soil • Surface water

A resource can be renewable in principle yet depleted in practice.

Water: Abundant on Earth, Limited for Human Use

~71%

Earth’s surface covered by water

Most water is saline or not readily available

~0.5%

commonly cited usable & available freshwater estimate

Planetary water abundance ≠ freshwater availability where and when people need it.

Teaching note: ~0.5% is an approximate communication statistic; definitions vary by accounting method.

India’s Water Reality

18%
world population

~4%
global water resources

~50%
decline in per-capita availability since 1970

Population › **Agriculture** › **Urbanization** › **Industry** › **Groundwater** › **Climate variability**

Civil Engineering challenge: reliable water services with lower losses, over-extraction and inefficiency.

Source basis: World Bank India water-sector evidence / 2026 brief.

Per-Capita Water Availability: India in Context

A better way to understand water pressure is to ask:

How much renewable water is available per person each year?

INDIA

~1,014

m^3/person/year

2022 • renewable internal freshwater resources

WORLD

~5,393

m^3/person/year

2022 • same international dataset

INDIA ≈ 19%

of the global average

≈ 81% lower than the world average

internationally comparable renewable freshwater measure

Source: FAO AQUASTAT via World Bank; data year 2022.

India’s Water Stress Threshold

India’s national assessment uses annual per-capita water availability as an indicator of water stress.



2021 India average annual availability

1,486 m^3/person/year

→ below the 1,700 m^3/person/year stress threshold

2031 projection

1,367 m^3/person/year

→ continued water-stressed range

Source: Ministry of Jal Shakti, Government of India. Thresholds are broad indicators of water stress/scarcity.

Why Per-Capita Water Availability Matters to Engineers

Per-capita availability is not the same as water actually delivered to a user.



A useful engineering interpretation

Population growth can reduce water available per person even when the total renewable resource changes little.

Civil Engineering response

- Reduce distribution losses
- Improve water-use efficiency
- Harvest and recharge rainwater
- Reuse treated wastewater
- Protect source and watershed
- Manage demand, not only supply

Population growth converts a fixed or slowly changing water resource into declining per-person availability.

Resource Potential ≠ Resource Availability

Natural Potential → Technically Accessible → Economically Feasible → Right Location → Right Time → Usable Resource

sunlight ≠ electricity • rainfall ≠ drinking water

mineral deposit ≠ construction material • groundwater ≠ sustainable water supply

Engineering converts resource potential into useful services — subject to technology, cost, location, time and ecological constraints.

Solar Energy: An Abundant Renewable Resource

~748 GW

estimated solar potential for India under stated assessment assumptions

Why potential does not become capacity automatically

PV Technology → Land → Materials → Storage → Transmission → Grid + Investment

An abundant renewable resource can still be constrained by infrastructure, materials, storage, land and economics.

Source basis: MNRE/NISE solar-potential assessment; verify assessment year/assumptions when presenting the number.

What Is Resource Depletion?

Resource depletion occurs when extraction / consumption persistently exceeds natural replenishment or practical recovery.

- Over-Extraction
- Over-Consumption
- Pollution
- Land Degradation
- Loss of Regenerative Capacity

Groundwater depletion • Forest loss • Soil degradation • Mineral exhaustion • Fossil-fuel depletion

A resource need not disappear completely to become critically depleted.

Groundwater: A Civil Engineering Example

Recharge → Aquifer Storage → Natural Discharge

Human extraction ↓

Recharge > extraction → storage increases
Recharge ≈ extraction → balance / stress
Recharge < extraction → water-table decline

Urbanization → impervious surfaces → reduced infiltration

Population growth → increased demand → increased extraction

Reduced recharge + increased extraction = groundwater stress.

Land, Soil and Minerals: The Slow Resources

LAND

Natural / agricultural land → buildings, roads, industry and utilities.
Changes infiltration, runoff, habitat and local climate.

SOIL

Can regenerate, but formation is extremely slow compared with erosion and degradation.

MINERALS

Iron • Copper • Aluminium • Limestone
Geological formation makes stocks effectively finite.

Some resources are renewable in principle but effectively non-renewable at human timescale.

Resource	Typical natural timescale	Engineering interpretation
Land	Essentially fixed in area	Once urbanized/sealed, restoration to original ecological function may take decades or may be impractical
Topsoil	Hundreds–thousands of years to develop substantially	Can be eroded or removed in hours/days, but replacement is extremely slow
Minerals	Millions of years through geological processes	Extraction occurs vastly faster than geological formation
Fossil fuels	Millions of years	Consumed within decades/centuries → effectively non-renewable

What Is Carrying Capacity?

Maximum level of population, activity or resource use an environment can sustain over time without unacceptable degradation of essential functions.

Below capacity › **Near capacity** › **Beyond capacity**

Examples: City water supply • River assimilative capacity • Landfill handling capacity

Capacity is about sustained function — not simply a single engineering design number.

Carrying Capacity Is Dynamic

Effective capacity depends on

Resource Availability › Technology › Infrastructure › Consumption Pattern › Sensitivity › Governance

Same population • different resource demand

CITY A

High leakage • inefficient buildings • private vehicles • low wastewater reuse

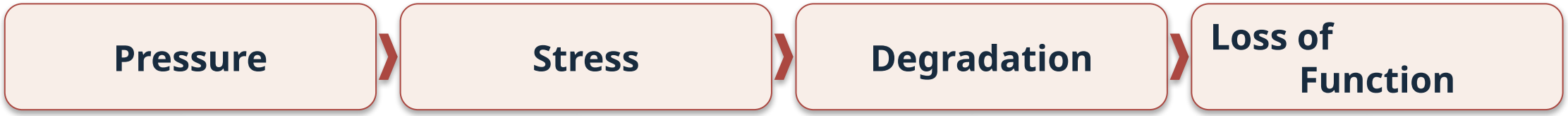
CITY B

Low leakage • Efficient buildings • Public transport • Wastewater reuse

Better engineering and management can increase effective capacity — but cannot eliminate ecological limits.

Environmental Stress: When the System Comes Under Pressure

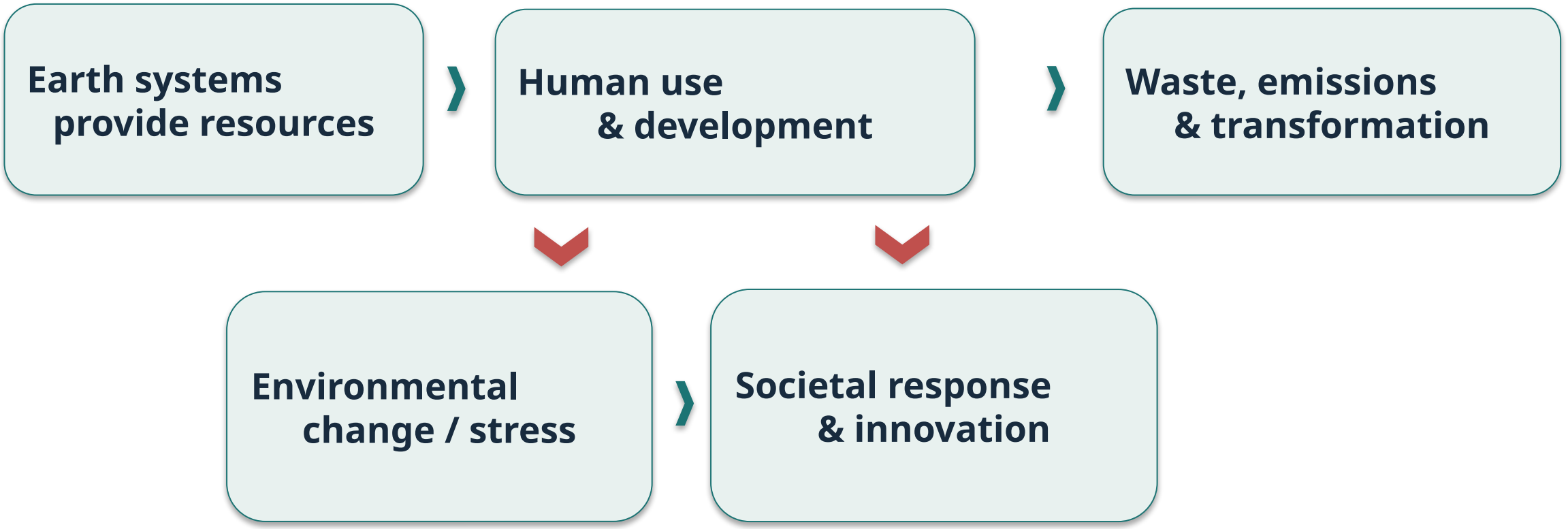
Pressure on an environmental system that reduces its ability to maintain normal functions.



WATER: over-extraction • pollution • altered flows
LAND: erosion • urbanization • contamination
AIR: emissions • particulate pollution

ECOSYSTEMS: habitat loss • fragmentation • resource overuse

Human–Environment Feedback Loop



Humans are not separate from the environment; humans are active components of the Earth system.

The Sustainability Challenge: The Engineer's Choice

CONVENTIONAL PATHWAY

More Demand → More Extraction → More Infrastructure → More Resource Consumption → More Pressure

SUSTAINABLE PATHWAY

Understand Limits → Reduce Demand → Improve Efficiency → Reuse / Recover → Protect Functions

"Sustainable engineering is not about stopping development. It is about changing the way development uses resources."

Final question: Can we meet human needs without exceeding the capacity of the Earth systems that support us?